Intelligent Agents

Assignment: Designing and Evaluating an Intelligent Agent

Chosen Agent: AI Patient Follow-Up and Care Management Agent

After receiving treatment, patients are frequently left to remember their own follow-up appointments, tests, and medication schedules — and many simply forget. This assignment presents the design and evaluation of an intelligent agent — the **AI Patient Follow-Up and Care Management Agent** — that monitors a patient's care schedule and provides timely reminders and follow-up support. It covers problem identification, problem formulation, PEAS-based evaluation, and classification of the agent's environmental properties, following the standard framework used for analyzing intelligent agents.

1. Identify the Problem

Once a patient leaves a hospital or clinic, responsibility for remembering follow-up care usually falls entirely on the patient themselves. In practice, this creates several recurring problems:

- **Missed follow-up visits** — patients forget to return for check-ups after surgery, treatment, or diagnosis.
- **Skipped tests** — lab tests or scans that were advised are delayed or never completed.
- **Inconsistent medication use** — patients forget doses or stop medication earlier than prescribed, without a system to track adherence.

These gaps can worsen health outcomes and increase the burden on the healthcare system through avoidable complications and re-admissions.

Problem Statement

> *There is a need for an intelligent agent that can monitor a patient's care schedule — including follow-up appointments, tests, and medication timings — and provide timely reminders and appropriate follow-up support, so that patients stay on track with their care plan without relying solely on memory.*

2. Formulate the Problem (Text-Based)

The problem is formulated below in terms of the agent's initial state, available actions, goal state, percepts, and the sequence of actions it follows.

Initial State

The agent has no knowledge of a patient's care plan until it is entered into the system by the hospital or clinic — for example, the discharge date, follow-up date, prescribed tests, and medication schedule.

Actions Available to the Agent

- Send a reminder notification for an upcoming appointment or test
- Send a medication reminder at the scheduled time
- Record the patient's response or confirmation
- Escalate an alert to a caregiver or doctor if a reminder is repeatedly ignored
- Update the patient's care record

Goal State

The patient completes all scheduled follow-up appointments, tests, and medication doses on time, with missed items flagged early enough for a caregiver or doctor to intervene.

Percepts (Inputs the Agent Receives)

- Follow-up appointment date and time
- Medication schedule (dose and timing)
- Test or scan due dates
- Patient's confirmation or response to a reminder

Sequence of Actions (Path)

Retrieve patient's care plan → Monitor upcoming dates and doses → Send reminder at the right time → Track patient response → Escalate to caregiver/doctor if missed → Update record → Repeat

Sample Scenario (Illustrative Data)

Field	Value
Patient Name	Bilal
Procedure	Minor Surgery (Appendix Removal)
Follow-Up Date	10 days after discharge
Medication Schedule	Antibiotic — twice daily for 7 days
Reminder Channel	SMS and mobile app notification

Formulated Problem Statement

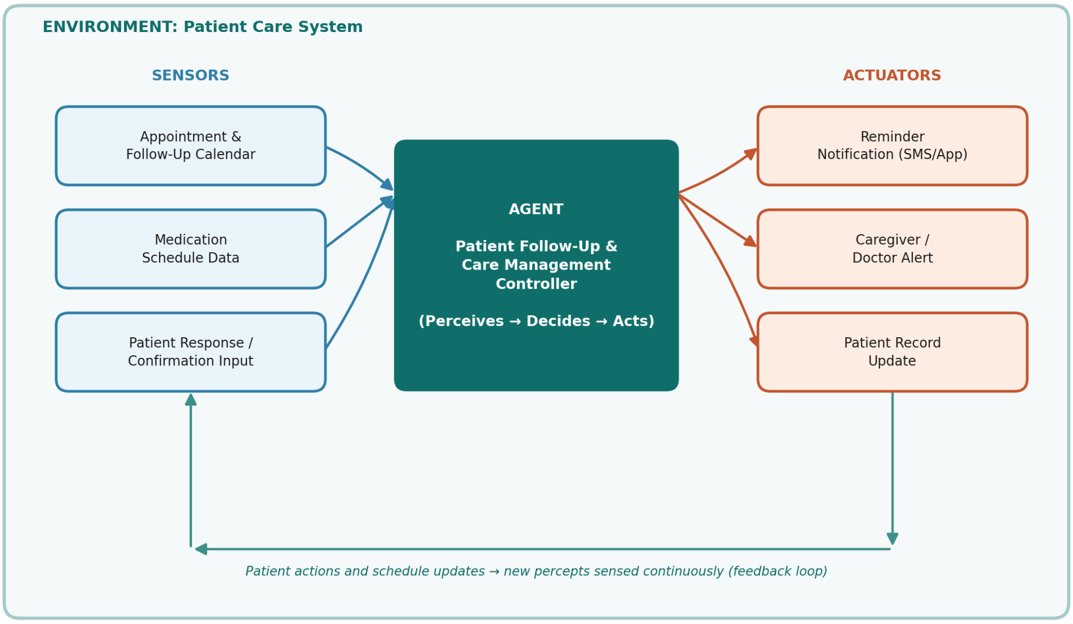
> *Given a patient's follow-up appointment dates, medication schedule, and test due dates as input, the agent must send timely reminders and track responses, such that the patient completes their care plan on schedule, with unresolved reminders escalated to a caregiver or doctor.*

2.1 Graphical Representation of the Environment

The diagram below illustrates the agent's perception–action cycle within the patient care environment: sensors gather the patient's schedule and responses, the agent processes this information and decides on the right action, and actuators send reminders or alerts back into the environment — where the patient's actions and schedule updates produce new percepts for the next cycle.

Graphical Environment Representation

AI Patient Follow-Up and Care Management Agent — Perception–Action Cycle



3. Evaluate in the Context of PEAS

PEAS defines the task environment of an agent through four components: **P**erformance measure, **E**nvironment, **A**ctuators, and **S**ensors.

Component	Details
P — Performance Measure	Percentage of appointments and tests completed on time, medication adherence rate, reduction in missed follow-ups, and overall patient satisfaction.
E — Environment	Patients, hospital/clinic scheduling system, caregivers, and doctors.
A — Actuators	Reminder notification (SMS/app), caregiver or doctor alert, and patient record update.
S — Sensors	Appointment and follow-up calendar, medication schedule data, and patient response/confirmation input.

4. Identify the Environmental Properties

The agent's environment is classified below using the standard set of environment properties.

Property	Type	Justification
Observability	Partially Observable	The agent only knows what is logged in the system or confirmed by the patient — it cannot directly observe whether a medication dose was actually taken, so its knowledge stays incomplete.

Property	Type	Justification
Determinism	Stochastic	Sending a reminder does not guarantee the patient will respond or follow through — human behaviour introduces uncertainty into the outcome.
Episodes	Sequential	A missed reminder or appointment affects how future reminders and escalations are handled, so decisions are linked across time rather than independent.
Change	Dynamic	Appointments get rescheduled, new tests get added, and patient responses arrive at any time, independently of when the agent is processing a decision.
Values	Discrete	Appointments, medication doses, and reminders occur as distinct, countable events (e.g., a specific dose at a specific time) rather than continuously varying quantities.
Agents	Multi-Agent	Multiple patients are monitored at once, and caregivers or doctors also act within the same system, making it a multi-agent environment.

End of Assignment — AI Patient Follow-Up and Care Management Agent